



Original Investigation | Anesthesiology

Efficacy and Safety of Esketamine for Supplemental Analgesia During Elective Cesarean Delivery

A Randomized Clinical Trial

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Abstract

IMPORTANCE Epidural anesthesia is a primary choice for cesarean delivery, but supplemental analgesics are often required to relieve pain during uterine traction.

OBJECTIVE To investigate the sedative and analgesic effects of intravenous esketamine administered before childbirth via cesarean delivery with the patient under epidural anesthesia.

DESIGN, SETTING, AND PARTICIPANTS This multicenter, double-blind randomized clinical trial assessed 903 women 18 years or older who had full-term single pregnancy and were scheduled for elective cesarean delivery with epidural anesthesia in 5 medical centers in China from September 18, 2021, to September 20, 2022.

INTERVENTION Patients were randomized to receive intravenous injection of 0.25 mg/kg of esketamine or placebo before incision.

MAIN OUTCOMES AND MEASURES The coprimary outcomes included scores on the numeric rating scale of pain (an 11-point scale, with 0 indicating no pain and 10 indicating the worst pain; a difference of ≥ 1.65 points was clinically meaningful) and Ramsay Sedation Scale (a 6-point scale, with 1 indicating restlessness and 6 indicating deep sleep without response; a difference of ≥ 2 points was clinically meaningful) immediately after fetal delivery. Secondary outcomes included neonatal Apgar score assessed at 1 and 5 minutes after birth.

RESULTS A total of 600 women (mean [SD] age, 30.7 [4.3] years) were enrolled and randomized; all were included in the intention-to-treat analysis. Immediately after fetal delivery, the score on the numeric rating scale of pain was lower with esketamine (median [IQR], 0 [0-1]) than with placebo (median [IQR], 0 [0-2]; median difference, 0; 95% CI, 0-0; $P = .001$), but the difference was not clinically important. The Ramsay Sedation Scale scores were higher (sedation deeper) with esketamine (median [IQR], 4 [3-4]) than with placebo (median [IQR], 2 [2-2]; median difference, 2; 95% CI, 2-2; $P < .001$). The neonatal Apgar scores did not differ between the 2 groups at 1 minute (median difference, 0; 95% CI, 0-0; $P = .98$) and at 5 minutes (median difference, 0; 95% CI, 0-0; $P = .27$). Transient neurologic or mental symptoms were more common in patients given esketamine (97.7% [293 of 300]) than in those given placebo (4.7% [14 of 300]; $P < .001$).

CONCLUSIONS AND RELEVANCE For women undergoing cesarean delivery under epidural anesthesia, a subanesthetic dose of esketamine administered before incision produced transient analgesia and sedation but did not induce significant neonatal depression. Mental symptoms and nystagmus were common but transient. Indications and the optimal dose of esketamine in this

(continued)

Key Points

Question Is intravenous esketamine safe and effective before childbirth during cesarean delivery with epidural anesthesia?

Findings In this multicenter, double-blind, randomized clinical trial of 600 women undergoing cesarean delivery with epidural anesthesia, a subanesthetic dose of esketamine administered before incision produced transient analgesia and sedation and easily passed through the placental barrier but did not induce significant neonatal depression. Mental symptoms and nystagmus were common but transient.

Meaning Indications and the optimal dose of esketamine in this patient population need further clarification, but study should be limited to those who require supplemental analgesia.

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Abstract (continued)

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Introduction

Epidural anesthesia is a widely used technique for cesarean delivery.¹ For example, it is the first-choice method when labor epidural analgesia is converted to cesarean delivery anesthesia² and when it is critical to avoid hypotension during induction of neuraxial blockage.^{3,4} However, because of incomplete block of the splanchnic nerve, some parturients experience discomfort or pain during exploration of the uterus, pulling peritoneum, and fetal delivery.⁵ The referred pain not only brings painful memories and mental stress to parturients but also affects the conduct of the surgical procedure. In some cases, change to general anesthesia is required.⁶ Therefore, it is often necessary to administer supplemental analgesics during cesarean delivery with epidural anesthesia.⁶ Opioids (such as fentanyl and sufentanil) and dexmedetomidine have been administered for this purpose; however, adverse effects, including respiratory depression, itching, and hypotension, limit their widespread use.⁷

Esketamine, a dextrorotatory isomer of ketamine, is 2 times more potent than ketamine in hypnotic and analgesic effect and less likely to produce psychiatric adverse reactions.^{8,9} In a preclinical study of pregnant sheep, Strümper et al¹⁰ verified that esketamine has similar effects on uterine perfusion and has limited effects on maternal and fetal hemodynamics and respiration when compared with racemic ketamine; thus, it might be an appropriate analgesic in the obstetric setting. In a clinical study of women undergoing cesarean delivery, Unlugenc et al¹¹ found that adding esketamine (0.05 mg/kg) to intrathecal plain bupivacaine (10 mg) resulted in rapid onset of sensory and motor block and enhanced segmental spread of spinal block but did not prolong the duration of spinal anesthesia. Suppa et al¹² reported that, for women after cesarean delivery, preventive esketamine administration (0.02-mg/kg intravenous injection of midazolam and 0.5-mg/kg intramuscular injection of esketamine at 10 minutes after childbirth followed by a 2-μg/kg/min intravenous infusion for 12 hours) had analgesic effects and was safe. However, the efficacy and safety of intravenous esketamine before childbirth are still unclear. We therefore investigated the sedative and analgesic effects of intravenous esketamine before childbirth during cesarean delivery with epidural anesthesia. We also observed placental transfer as well as the effects on maternal and fetal outcomes.

Methods

Study Design and Patient Enrollment

This double-blinded, placebo-controlled randomized clinical trial assessed 903 parturients from 5 medical centers for eligibility. The trial protocol was approved by the Clinical Research Ethics Committee of Women's Hospital, Zhejiang University School of Medicine and registered at ClinicalTrials.gov on September 16, 2021, and patients were enrolled between September 18, 2021, and September 20, 2022. All participants gave written informed consent. The formal trial protocol and statistical analysis plan are provided as [Supplement 1](#). There were no important changes after trial commencement. This report follows the Consolidated Standards of Reporting Trials ([CONSORT](#)) reporting guideline for randomized trials.

We enrolled women 18 years or older who had a full-term single pregnancy and regular prenatal examination and were scheduled for elective cesarean delivery with epidural anesthesia. The exclusion criteria included the following: (1) body mass index (calculated as weight in kilograms divided by height in meters squared) of 27 or greater; (2) previous mental illness, central nervous system disease, liver disease, abnormal kidney function, abnormal heart and lung function, diabetes, or American Society of Anesthesiologists (ASA) classification III or above; (3) severe obstetric complications, such as preeclampsia and eclampsia, pregnancy-induced hypertension, placenta previa, or placental abruption; (4) stillbirth or neonatal malformation; and (5) contraindications to epidural anesthesia, including abnormal coagulation function, anticoagulant therapy, severe hypovolemia, or hemodynamic instability.

Randomization and Blinding

Random allocations were generated using SPSS software, version 22.0 (IBM Corp) in a 1:1 ratio. Assignments were concealed in sequentially numbered opaque envelopes. Before anesthesia, study coordinators who otherwise were not involved in the trial opened the envelopes according to the recruitment sequence, prepared esketamine or an identical-appearing saline placebo, and gave the study drugs to the attending anesthesiologists (C.W., S.Q.Z., Y.B.P., Y.Q.H., and C.H.L.). All patients, anesthesiologists, other health care team members, and investigators who were responsible for data collection and follow-ups were blinded to group allocation.

Anesthesia Management and Intervention

No premedication was administered. In the operating room, heart rate and oxygen saturation were monitored continuously, and blood pressure was noninvasively measured every 5 minutes. Epidural catheterization was performed with parturients in the left lateral decubitus position. A 16-G Tuohy needle was used for epidural puncture in the L2-L3 intervertebral space. After confirmation of the epidural space by the loss of resistance to air method, a catheter was inserted 3 to 4 cm in a cephalad direction. Parturients were returned to the supine position with a 15° tilt to the left. Oxygen was provided at 5 L/min via a face mask.

A test dose of 5 mL of 1.5% lidocaine was given. After a 5-minute observation, 10 mL of 0.75% ropivacaine was given, followed by an infusion of 0.75% ropivacaine at 3 to 5 mL/h. The upper sensory block was targeted between the T4 and T6 levels. If not achieved, a supplemental dose of 5 mL of 0.75% ropivacaine was given. Study drugs were administered when the targeted upper sensory block level was achieved (approximately 2 minutes before incision). Specifically, 0.25 mg/kg of esketamine was administered intravenously for 1 minute for patients in the esketamine group; the same volume of normal saline was injected for patients in the control group. During epidural anesthesia, fluid infusion was administered per routine. Blood pressure was maintained within 20% from baseline. At the end of surgery, a patient-controlled epidural analgesia device was attached for postoperative analgesia, which was established with 0.2% ropivacaine and programmed to deliver a continuous infusion at 2 mL/h.

Data Collection and Outcome Assessment

Baseline data included demographic characteristics, pregestational comorbidity, number of gravidity and parturitions, duration of gestation, and ASA classification. Intraoperative data included durations of anesthesia and surgery, fluid balance, use of vasopressors, intervals from study drug administration and uterine incision to delivery, and neonatal data. Intervals from study drug administration to neonatal delivery and from uterine incision to neonatal delivery were also recorded.

Our coprimary outcomes included maternal pain intensity and sedation score immediately after fetal delivery. Pain intensity was assessed with the numeric rating scale (NRS; an 11-point scale, with 0 indicating no pain and 10 indicating the worst pain) at the following time points: before anesthesia, immediately after anesthesia, surgical incision (approximately 2 minutes after study drug

administration), 5 minutes after study drug administration, immediately after fetal delivery (approximately 10 minutes after esketamine), end of surgery, 6 hours after surgery, and 12 hours after surgery. A change of 1.65 points or more on the NRS was considered clinically meaningful.¹³ Maternal sedation level was assessed with the Ramsay Sedation Scale (with 1 indicating restless; 2, completely awake, quiet, and cooperative; 3, drowsy but responding to verbal commands; 4, lightly asleep but responding to touch or pain; 5, asleep but slowly responding to touch or pain; and 6, deeply asleep and does not respond) at the same time points as above. A change of 2 points or more against the Richmond Agitation-Sedation Scale was considered clinically meaningful.¹⁴

Our secondary outcomes included maternal pain intensity and sedation score at other time points, neonatal Apgar score assessed at 1 and 5 minutes after birth, and postnatal umbilical vein blood gas pH value. Among other outcomes, mean blood pressure, heart rate, and pulse oxygen saturation were measured before anesthesia, immediately after anesthesia, at incision (2 minutes after study drug administration), 5 minutes after study drug administration, immediately after fetal delivery (10 minutes after study drug administration), at end of surgery, and 1 hour after surgery. The requirement of neonatal ward admission and length of hospital stay after surgery were recorded.

After childbirth, samples of maternal arterial blood, neonatal umbilical arterial blood, and umbilical venous blood were collected in the esketamine group ($n = 13$) and stored in a -20°C refrigerator. Esketamine concentrations were measured with reverse-phase high-performance liquid chromatography.¹⁵

Adverse events were monitored. Specifically, hypotension was defined as a systolic blood pressure decrease less than 20% of baseline, hypertension was defined as a systolic blood pressure increase greater than 20% of baseline, bradycardia was defined as a heart rate less than 60 beats per minute, tachycardia was defined as a heart rate greater than 100 beats per minute, and desaturation was defined as oxygen saturation less than 90%. We also monitored the occurrence of nausea and vomiting, as well as neurologic and mental signs and symptoms (such as lethargy, diplopia, nystagmus, dizziness, headache, nightmare, hallucination, anxiety, and irritability). Adverse events were managed according to routine practice.

Statistical Analysis

On the basis of our pilot study, we hypothesized that the mean difference in NRS pain score immediately after fetal delivery (approximately 10 minutes after administering esketamine) would be 0.3, with SDs of 0.85 in the esketamine group and 1.23 in the control group. With α set at .05 and power at 90%, 263 patients were needed in each group to detect a significant difference. Considering a dropout rate of approximately 10%, we planned to enroll 300 patients in each group. The sample size was estimated with the PASS software, version 15.0 (NCSS).

Outcome analysis was performed in the intention-to-treat population; that is, all patients were analyzed in the group to which they were assigned. For the primary outcomes, we also performed analysis in the per-protocol population, excluding those with protocol deviation.

For continuous variables, the Kolmogorov-Smirnov test was applied to evaluate the distribution. Variables with normal distribution were presented as means (SDs), and intergroup comparisons were performed using the 2-tailed, unpaired t test. Variables with nonnormal distribution were presented as medians (IQRs), and intergroup comparisons were performed using the Mann-Whitney U test. Categorical variables were analyzed using the χ^2 test and are presented as number (percentage).

For the primary outcomes, the differences between groups were expressed as median differences (95% CIs). For other outcomes, the between-group difference of neonatal Apgar score was expressed as median difference (95% CI), that of umbilical vein pH as mean difference (95% CI), that of neonatal ward admission proportion as relative risk (RR) (95% CI), and that of length of hospital stay after surgery as hazard ratio (95% CI). For the primary outcomes, a 2-sided $P < .025$ ($.05/2$) was considered statistically significant. For other outcomes, a 2-sided $P < .05$ was considered statistically significant unless otherwise indicated after Bonferroni correction. Statistical analyses

were performed using SPSS for Windows, version 22.0 (IBM Corp) and GraphPad Prism, version 5.0 (GraphPad Software Inc).

Results

Patients

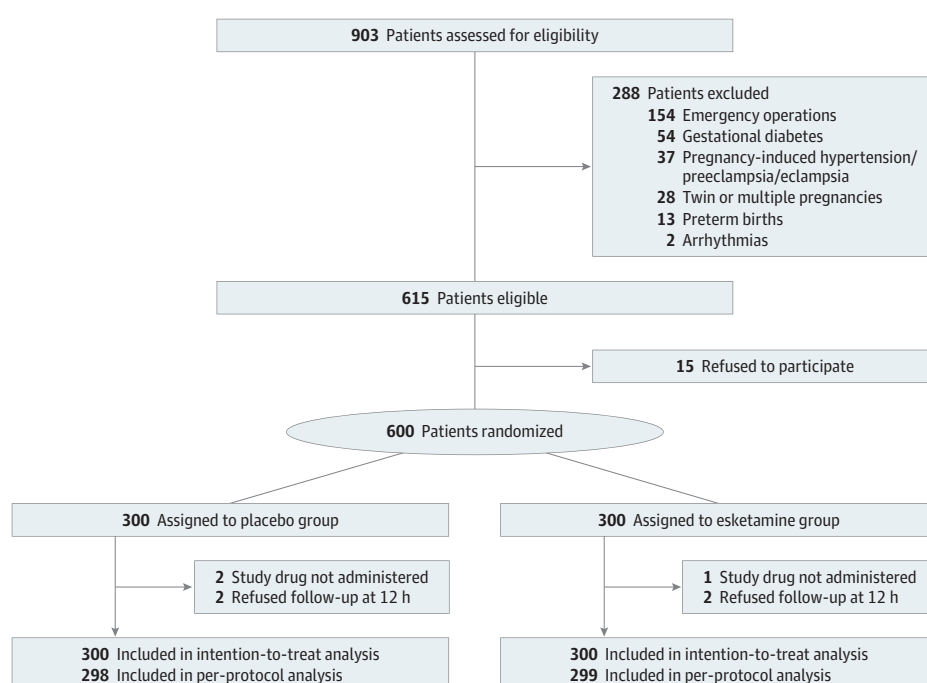
From 903 parturients who were assessed for eligibility, a total of 600 (mean [SD] age, 30.7 [4.3] years; mean [SD] prenatal body mass index, 25.9 [2.1]; mean [SD] gestational duration, 38.7 [1.1] weeks) were enrolled and randomized. During the study period, 3 women did not receive study drugs, and 4 women refused follow-up at 12 hours after the surgery. As a result, all 600 women ($n = 300$ in each group) were included in the intention-to-treat analysis, and 597 women were included in the per-protocol analysis (**Figure 1**). Baseline and intraoperative data, including the intervals from study drug administration to delivery, were comparable between the 2 groups (**Table 1**; eTable 1 in [Supplement 2](#)).

Efficacy Outcomes

Immediately after fetal delivery, the NRS pain score was lower in the esketamine group (median [IQR], 0 [0-1]) than in the placebo group (median [IQR], 0 [0-2]; median difference, 0; 95% CI, 0-0; $P = .001$); however, the difference was not clinically meaningful (**Figure 2A**). The Ramsay Sedation Score was higher (sedation deeper) in the esketamine group (median [IQR], 4 [3-4]) than in the placebo group (median [IQR], 2 [2-2]); median difference, 2; 95% CI, 2-2; $P < .001$); the difference was clinically important (**Figure 2B**). Per-protocol analysis gave similar results (**Table 2**).

Among secondary and other outcomes, the NRS pain score at other time points did not differ between groups (**Figure 2A**). The Ramsay Sedation Scores were higher (sedation deeper) in the esketamine group at surgical incision (median difference, 1; 95% CI, 1-1; $P < .001$) and at 5 minutes after study drug administration (median difference, 2; 95% CI, 2-2; $P < .001$) (**Figure 2B**; eTable 2 in [Supplement 2](#)). The neonatal Apgar scores at 1 and 5 minutes, the umbilical vein blood pH, and

Figure 1. Trial Diagram



hospital stay after surgery did not differ between the 2 groups. No neonatal ward admission was required in either group (Table 3). During the perinatal period, mean arterial pressure and heart rate were higher in the esketamine group than in the control group at 5 minutes after study drug

Table 1. Baseline Data and Intraoperative Data^a

Variable	Placebo group (n = 300)	Esketamine group (n = 300)
Baseline data		
Age, mean (SD), y	30.9 (4.3)	30.4 (4.2)
Height, mean (SD), cm	160.0 (5.3)	160.1 (5.5)
Weight, mean (SD), kg	66.4 (7.0)	66.9 (6.6)
Prenatal BMI, mean (SD)	25.9 (2.2)	25.9 (1.9)
Pregestational condition		
Medical comorbidity ^b	22 (7.3)	15 (5.0)
Gynecologic diseases ^c	13 (4.3)	11 (3.7)
History of surgery	17 (5.7)	20 (6.7)
No. of gravidities, median (IQR)	2 (1-2)	2 (1-3)
No. of parturitions, median (IQR)	0 (0-1)	0 (0-1)
Duration of gestation, mean (SD), wk	38.8 (1.2)	38.7 (1.2)
ASA classification II	300 (100)	300 (100)
Study site		
Site 1	90 (30.0)	90 (30.0)
Site 2	40 (13.3)	40 (13.3)
Site 3	80 (26.7)	80 (26.7)
Site 4	60 (20.0)	60 (20.0)
Site 5	30 (10.0)	30 (10.0)
Intraoperative data		
Duration of anesthesia, mean (SD), min	62 (16)	64 (17)
Fluid infusion, mean (SD), mL	1000 (201)	1009 (252)
Estimated blood loss, mean (SD), mL	290 (93)	299 (119)
Duration of surgery, mean (SD), min	49 (16)	51 (17)
Use of vasopressors ^d	163 (54.3)	157 (52.3)
Study drug injection to delivery, mean (SD), min	5.1 (2.8)	4.8 (2.8)
Uterine incision to delivery, mean (SD), min	1.8 (0.89)	1.7 (1.0)
Male neonate	146 (48.7)	155 (51.7)
Birth weight, mean (SD), g	3335 (499)	3335 (458)

Abbreviations: ASA, American Society of Anesthesiologists; BMI, body mass index (calculated as weight in kilograms divided by height in meters squared).

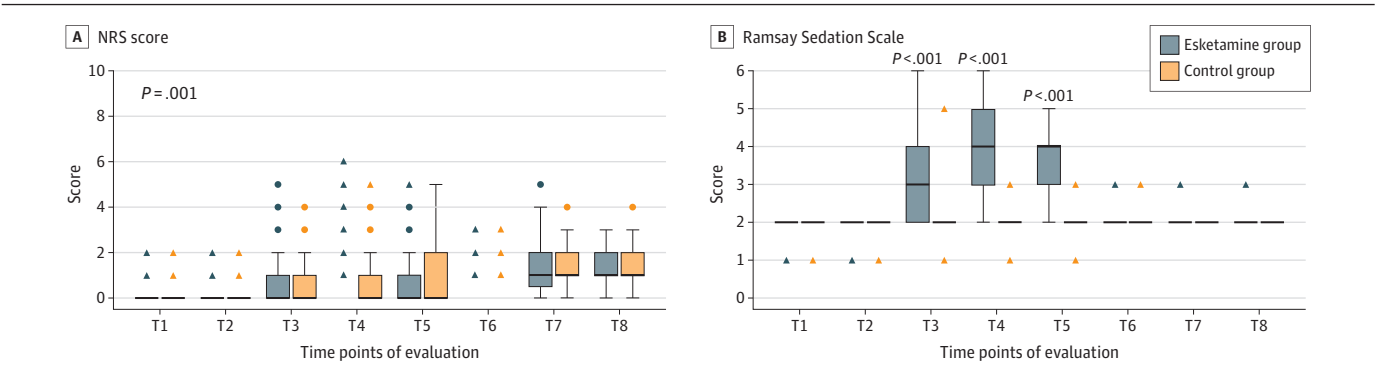
^a Data are presented as number (percentage) of patients unless otherwise indicated.

^b Including asthma, arrhythmia, latent glomerulonephritis, abnormal liver function, and positive hepatitis B surface antigen.

^c Including hysteromyoma, ovarian cysts, dysfunctional uterine bleeding, polycystic ovary syndrome, and pelvic inflammatory disease.

^d Included ephedrine and phenylephrine.

Figure 2. Comparison of the Numeric Rating Scale (NRS) Pain Scores and the Ramsay Sedation Scale Scores Between the 2 Study Groups



The NRS is an 11-point scale, with 0 indicating no pain and 10 indicating the worst pain. The Ramsay Sedation Scale is a 6-point scale, with 1 indicating restlessness; 2, completely awake, quiet, and cooperative; 3, drowsiness but responding to verbal commands; 4, light sleep but responding to touch or pain; 5, asleep but slowly responding to touch or pain; and 6, deeply asleep and does not respond. The box and whiskers plots show medians, IQRs, and outer ranges; individual points indicate mild outliers (circles, outside

1.5 times of the IQR) and extreme outliers (triangles, outside 3 times of interquartile range). T1 indicates before anesthesia; T2, immediately after anesthesia; T3, surgical incision (approximately 2 minutes after study drug administration); T4, 5 minutes after study drug administration; T5, immediately after fetal delivery (approximately 10 minutes after study drug administration); T6, end of surgery; T7, 6 hours after surgery; and T8, 12 hours after surgery.

administration and immediately after fetal delivery (eFigure, A and B in Supplement 2). There were no differences between groups regarding oxygen saturation at all time points (eFigure, C and eTable 3 in Supplement 2).

In an exploratory analysis, the proportion of patients with an NRS pain score greater than 2 immediately after fetal delivery was significantly lower in patients given esketamine (6.0% [18 of 300] with esketamine vs 10.7% [32 of 300] with placebo; RR, 0.56; 95% CI, 0.32-0.98; $P = .04$); the proportion of patients with Ramsay Sedation Scale scores of 4 or higher immediately after fetal delivery was significantly higher in patients given esketamine (71.7% [215 of 300]) vs placebo (0% [0 of 300]) (RR, 431.00; 95% CI, 27.00-6881.35; $P < .001$) (Table 2).

After childbirth, the mean (SD) plasma concentrations of esketamine were 172.4 (79.8) ng/mL in maternal arterial blood, 62.0 (23.6) ng/mL in neonatal umbilical arterial blood, and 142.9 (66.7) ng/mL in umbilical venous blood. The mean (SD) ratio of fetal umbilical vein to maternal artery was 0.83 (0.09), and the ratio of umbilical artery to umbilical vein was 0.48 (0.20) (eTable 4 in Supplement 2).

Safety Outcomes

During the intraoperative period, more patients given esketamine developed neurologic or mental symptoms (97.7% [293 of 300]) compared with the placebo group (4.7% [14 of 300]; $P < .001$), mainly somnolence (70.7% [212 of 300] vs 0.0% [0 of 300], $P < .001$), nystagmus (67.3% [202 of 300] vs 0.0% [0 of 300], $P < .001$), vertigo (44.7% [134 of 300] vs 0.3% [1 of 300], $P < .001$), dizziness (15.0% [45 of 300] vs 4.0% [12 of 300], $P < .001$), hypertonia (7.7% [23 of 300] vs 0.0% [0 of 300], $P < .001$), and daymare (1.3% [4 of 300] vs 0.0% [0 of 300], $P = .04$). Among other adverse events, fewer patients given esketamine developed hypotension (4.7% [14 of 300]) compared with the placebo group (8.7% [26 of 300], $P = .047$), but more developed hypertension

Table 2. Efficacy Outcomes

Outcome	Placebo group (n = 300)	Esketamine group (n = 300)	Estimated effects (95% CI) ^a	P value
Primary outcomes				
Numeric rating scale pain at rest score, median (IQR) ^b				
Immediately after fetal delivery	0 (0 to 2)	0 (0 to 1)	Median difference, 0 (0 to 0)	.001 ^c
Immediately after fetal delivery (per-protocol analysis)	0 (0 to 2) (n = 298)	0 (0 to 1) (n = 299)	Median difference, 0 (0 to 0)	.001 ^c
Ramsay Sedation Scale score, median (IQR) ^d				
Immediately after fetal delivery	2 (2 to 2)	4 (3 to 4)	Median difference, 2 (2 to 2)	<.001 ^c
Immediately after fetal delivery (per-protocol analysis)	2 (2 to 2) (n = 298)	4 (3 to 4) (n = 299)	Median difference, 2 (2 to 2)	<.001 ^c
Secondary outcomes				
Neonatal Apgar score, median (IQR)				
1 min	10 (10 to 10)	10 (10 to 10)	Median difference, 0 (0 to 0)	.98
5 min	10 (10 to 10)	10 (10 to 10)	Median difference, 0 (0 to 0)	.27
Umbilical vein pH, mean (SD)	7.35 (0.05)	7.34 (0.06)	Median difference, 0 (0 to 0)	.58
Other predefined outcomes				
Admission to neonatal ward, No. (%) ^e	0	0	NA	>.99
Length of hospital stay after surgery, mean (SD), d	4.2 (0.9)	4.2 (0.9)	Mean difference, 0.0 (-0.1 to 0.2)	.76
Exploratory analysis				
Numeric rating scale pain score >2 immediately after fetal delivery, No. (%) ^b	32 (10.7)	18 (6.0)	RR, 0.56 (0.32 to 0.98)	.04
Ramsay Sedation Scale score ≥4 immediately after fetal delivery, No. (%) ^d	0	215 (71.7)	RR, 431.00 (27.00 to 6881.35)	<.001

Abbreviations: NA, not applicable; RR, relative risk.

^a Calculated as the esketamine group minus or vs the control group.

^b Score ranges from 0 to 10, with 0 indicating no pain and 10 indicating the worst pain.

^c P value considered statistically significant after Bonferroni correction.

^d Ramsay Sedation Score ranges from 1 (restlessness) to 6 (deeply asleep and does not respond), with 2 indicating completely awake, quiet, and cooperative.

^e For the purpose of further monitoring and/or treatment that was considered necessary by the pediatricians. Indications included hypoglycemia, neonatal malformation, neonatal vomiting, and premature delivery.

(7.7% [23 of 300] vs 1.3% [4 of 300], $P < .001$) and tachycardia (27.0% [81 of 300] vs 16.7% [50 of 300], $P = .002$). During the first postoperative day, more patients given esketamine developed nausea or vomiting (23.0% [69 of 300]) compared with the placebo group (11.0% [33 of 300], $P < .001$), but mental symptoms did not differ between the groups (Table 3).

Discussion

Results of this multicenter randomized clinical trial showed that, for women undergoing elective cesarean delivery with epidural anesthesia, a single subanesthetic dose of esketamine administered before incision transiently relieved pain intensity, but the effect was not clinically important; in the meantime, it produced transient but clinically important sedation. A significant part of esketamine transferred across the placenta to the fetus, but the neonatal Apgar score was not affected. Use of esketamine significantly increased neurologic and mental symptoms, but the adverse effects were transient and only appeared during the intraoperative period.

Neuraxial anesthesia, including epidural anesthesia, is the primary choice for cesarean delivery, but supplemental analgesics are often required to relieve visceral traction pain during surgery.^{16,17} Ketamine is a frequently used analgesic for this purpose.¹⁸ However, the potential effects of neonatal depression and even neuro-toxic effects, as well as sympathetic excitation and psychiatric symptoms,^{19,20} raised concern when using ketamine.^{21,22} Esketamine is more potent in analgesia and less likely to produce psychiatric adverse effects^{8,9}; thus, esketamine might be a suitable substitute for women undergoing cesarean delivery. Furthermore, both ketamine and esketamine are proven to

Table 3. Adverse Events in the Study Participants

Adverse event	Placebo group, No. (%) (n = 300)	Esketamine group, No. (%) (n = 300)	P value
Intraoperative period			
Hypotension ^a	26 (8.7)	14 (4.7)	.046
Hypertension ^b	4 (1.3)	23 (7.7)	<.001
Bradycardia ^c	2 (0.7)	0	.16
Tachycardia ^d	50 (16.7)	81 (27.0)	.002
Desaturation ^e	0	0	NA
Nausea or vomiting	48 (16)	52 (17.3)	.66
Neurologic or mental symptoms	14 (4.7)	293 (97.7)	<.001
Somnolence	0	212 (70.7)	<.001
Nystagmus	0	202 (67.3)	<.001
Vertigo	1 (0.3)	134 (44.7)	<.001
Dizziness	12 (4.0)	45 (15.0)	<.001
Hypertonia	0	23 (7.7)	<.001
Daymare	0	4 (1.3)	.04
Hallucination	1 (0.3)	3 (1.0)	.32
Diplopia	0	0	NA
Postoperative period			
Desaturation ^e	0	0	NA
Nausea or vomiting	33 (11.0)	69 (23.0)	<.001
Neurologic or mental symptoms	0	1 (0.3)	>.99
Somnolence	0	0	NA
Nystagmus	0	0	NA
Vertigo	0	0	NA
Dizziness	0	0	NA
Hypertonia	0	0	NA
Daymare	0	0	NA
Hallucination	0	0	NA
Diplopia	0	1 (0.3)	>.99

Abbreviation: NA, not applicable.

^a Defined as systolic blood pressure less than 90 mm Hg or a decrease of greater than 30% from baseline.

^b Defined as systolic blood pressure greater than 160 mm Hg or an increase of greater than 30% from baseline.

^c Defined as heart rate less than 60 beats per minute or a decrease of greater than 20% from baseline.

^d Defined as heart rate greater than 100 beats per minute or an increase of greater than 20% from baseline.

^e Defined as pulse oxygen saturation less than 90% or a decrease of more than 5% (absolute value) from baseline.

have antidepressant effects.²³ In fact, low-dose ketamine and esketamine have been used prophylactically in women undergoing cesarean delivery and are reported to be effective in improving analgesia and relieving depression.^{1,12,24-26} In a sequential dose-finding study, an intravenous dose of 0.15 mg/kg of esketamine injected before surgery significantly reduced the 95% effective dose of spinal ropivacaine for cesarean delivery.²⁷ We therefore adopted a prophylactic regimen.

In the current study, an intravenous low dose (0.25 mg/kg) of esketamine was injected before incision. Different from previous results with ketamine,¹⁸ the results of our study did not indicate clinically important improvement in analgesia either intraoperatively or up to 12 hours after surgery. For example, in a previous study,²⁸ an intravenous dose of ketamine (0.25 mg/kg) administered before incision significantly improved 24-hour analgesia after cesarean delivery with spinal anesthesia. This finding was possibly attributable to the perfect analgesia in our control group patients, because the sensory block level reached T6 or above before incision and epidural analgesia was provided after surgery. As a result, the median NRS pain score was maintained between 0 and 1 during the evaluation period; thus, there is no space to further improve analgesia in our patients. Despite this, the proportion of our patients with NRS pain scores of 3 or higher was significantly lower with esketamine. Low-dose esketamine should thus be reserved for patients who actually experience pain. In line with other studies,^{29,30} our study found that a single low dose of esketamine produced clinically significant sedation in our patients, but the effect was transient and did not persist into the postoperative hours.

Maternal hypotension frequently occurs during epidural anesthesia for cesarean delivery and, if severe, will lead to adverse maternal or fetal outcomes. Esketamine can stimulate the sympathetic nervous system and increase blood pressure and heart rate, especially when given in high doses.³¹ In the study by Zhang et al,²⁷ use of low-dose esketamine (0.15 mg/kg) was associated with a decreased incidence of hypotension during spinal anesthesia for cesarean delivery. It is therefore recommended that ketamine and esketamine should be considered for women undergoing elective cesarean delivery and at a high risk of hypotension.²⁷ In accordance with these results, we also found that patients given esketamine had higher mean arterial blood pressure and heart rate, but again, the effects were transient and only significant at 5 minutes after study drug administration and immediately after fetal delivery.

Ketamine was approved for clinical anesthesia in 1970.³² For women undergoing cesarean delivery, ketamine or esketamine has been used for induction of general anesthesia,³¹ as an adjuvant for neuraxial anesthesia,^{11,27} or for postoperative analgesia.³³ Placental transfer is always a concern when administering ketamine before childbirth. In the current study, a single intravenous dose (0.25 mg/kg) of esketamine was administered before incision. In our results, the mean (SD) umbilical vein to maternal artery ratio of esketamine concentration was 0.83 (0.09), which was consistent with previous studies^{34,35} and further demonstrated that esketamine can easily pass through the placental barrier and reach the fetus. We found a mean (SD) umbilical artery to vein drug concentration ratio of 0.48 (0.20), which suggests that esketamine might be rapidly metabolized or redistributed in the fetus, but further studies are required to clarify the observed results. Despite these findings, we did not find significant deterioration in neonatal outcomes, indicating that the adopted dose was relatively safe for neonates in this patient population. As expected, most of our patients developed mental symptoms and nystagmus, but the symptoms were transient and did not require treatment.

Limitations

There are some limitations to this study. First, because the epidural block level and analgesic effect were sufficient for our control group patients, we did not find clinically significant improvement in intraoperative and postoperative analgesia with intervention, which questions the necessity of administering esketamine in our patients. However, this ideal situation occurred in a strictly performed trial and could not exclude the requirement of supplemental analgesics in routine

practice. Second, we only tested the effects of a fixed dose of esketamine (0.25 mg/kg). A lower dose of esketamine might also be effective but with fewer adverse events. Further studies are required to clarify the indications and effects of supplemental esketamine in this patient population. Third, we did not evaluate participants' experience during the study period. The neurologic and mental symptoms provoked by esketamine might produce unfavorable experience and low patient satisfaction. Fourth, we only measured esketamine plasma concentrations in 13 mothers and their fetuses. Although our results revealed the degree of placental transfer of esketamine, they may not accurately represent the overall situation of the esketamine group.

Conclusions

The results of this randomized clinical trial showed that, for women undergoing cesarean delivery with epidural anesthesia, a subanesthetic dose of esketamine administered before incision did not have much effect on analgesia but raised concern about sedation and mental adverse effects. The administered esketamine easily passed through the placental barrier but did not induce significant neonatal depression. Indications and the optimal dose of esketamine in this patient population need further clarification, but study should be limited to those who require supplemental analgesia.

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SUPPLEMENT 1.

Trial Protocol

SUPPLEMENT 2.

- eTable 1. Enrollment by Study Site
- eTable 2. Numeric Rating Scale of Pain and Ramsay Sedation Scale During the Perinatal Period
- eTable 3. Blood Pressure, Heart Rate, and Pulse Oxygen Saturation During the Perinatal Period
- eTable 4. Maternal and Fetal Esketamine Concentrations
- eFigure. Comparison of the Mean Arterial Pressure (A), Heart Rate (B), and Pulse Oxygen Saturation (C) Between the Two Groups

SUPPLEMENT 3.

Data Sharing Statement

1 **Study Protocol and Statistical Analysis Plan**

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5 Supplement to: Li-Li Xu, Chun Wang, Chun-Mei Deng, et al. Esketamine for supplemental
6 analgesia during elective cesarean delivery: a randomized clinical trial

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8

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10 **This supplement contains the following items:**

11

12

13 1. Study protocol.

14 2. Statistical analysis plan.

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19

20

21 **Study protocol**

22

23

24

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27 Esketamine for supplemental analgesia during elective cesarean delivery: a randomized clinical
28 trial

29

30

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38 Affiliated Xiaoshan Hospital, Hangzhou Normal University.

39 Department of Anesthesiology, Jiaxing Maternity and Child Health Care Hospital.

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42 Version of protocol: V2.0

43 Date of version: July 30, 2020

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1. Background

Epidural anesthesia is widely used for cesarean delivery, especially in China. However, due to incomplete block of the splanchnic nerve, parturients often complain discomfort or pain during exploration of the uterus, pulling peritoneum, and fetal delivery. The referred pain brings painful memories and mental stress to parturients, and also affect the conduct of surgical procedure. In some cases, change to general anesthesia is required. Therefore, supplemental analgesics during cesarean delivery under epidural anesthesia is sometimes required and necessary.¹

Esketamine, a dextrorotatory isomer of ketamine, is two times more potent than ketamine in hypnotic and analgesic effect, with less psychiatric adverse reactions.^{2,3} In a pre-clinical study, Strümper et al.⁴ verified that esketamine has similar effects on uterine perfusion, and has limited effects on maternal/fetal hemodynamics and respiration when compared with racemic ketamine; it thus might be an appropriate analgesic in the obstetric setting. In clinical studies, Unlugenc et al.⁵ found that adding esketamine (0.05 mg/kg) to intrathecal plain bupivacaine (10 mg) for spinal anesthesia resulted in rapid onset of sensory and motor block and enhanced segmental spread of spinal block in patients undergoing cesarean delivery. Suppa et al.⁶ reported that, for women after cesarean delivery, preventive esketamine administration (0.5 mg/kg intramuscular injection at 10 minutes after childbirth, followed by a 2 µg/kg/min intravenous infusion for 12 h) has analgesic effects and was safe.

However, the efficacy and safety of intravenous esketamine before childbirth are still unclear. We therefore try to investigate the sedative and analgesic effects of intravenous esketamine before childbirth during cesarean delivery under epidural anesthesia. We will also explore the placental transfer of intravenous esketamine as well as the effects on maternal and fetal outcomes.

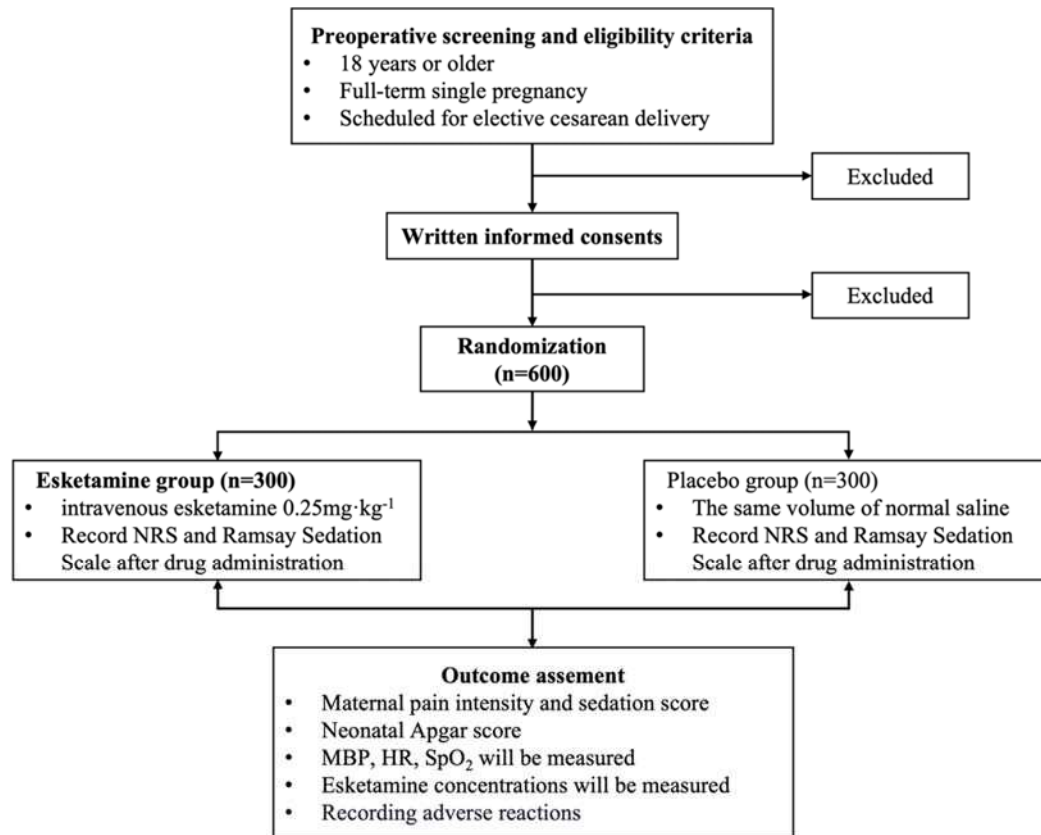
2. Purpose of the study

The purpose of the study is to investigate the sedative and analgesic effects of intravenous esketamine before childbirth during cesarean delivery under epidural anesthesia.

3. Study design

3.1 Type of the study

This is a multi-center, randomized, double blind trial with two parallel arms. The flow chart of the study is shown in Figure 1.



3.2 Sample size calculation

Based on a pilot study, we suppose that the mean difference in NRS pain score immediately after fetal delivery (10 min after administering esketamine) would be 0.3 with standard deviations (SD) of 0.85 and 1.23, respectively, between the two groups. With alpha set at 5% and power at 90%, 263 patients will be needed in each group. Considering a drop-out rate of about 10%, we plan to enroll 300 patients in each group. The sample size was estimated with the PASS 15.0 software.

3.3 Participating centers

3.3.1 This multi-center trial will be conducted in five hospitals. The five participating centers include Women's Hospital of Zhejiang University School of Medicine, Jiangxi Maternal and Child Health Hospital, Hangzhou Women's Hospital, Jiaxin Maternity and Child Health Care Hospital, and Xiaoshan Hospital.

3.3.2 The study is coordinated and monitored by the Department of Anesthesiology of Women's Hospital Zhejiang University School of Medicine; the Department of Anesthesiology and Critical Care Medicine of Peking University First Hospital is responsible for data management and data analysis.

4. Study participants

Potential participants will be screened before surgery by the qualified investigators.

4.1 Inclusion criteria

4.1.1 Women aged 18 years or older.

4.1.2 Full-term single pregnancy.

4.1.3 Scheduling for elective caesarean delivery.

4.1.4 Planning for epidural anesthesia.

4.1.5 Agree to participate, and give signed written informed consents

4.2 Exclusion criteria

Patients will be excluded if they meet any of the following criteria:

4.2.1 Body mass index (BMI) ≥ 27 kg/m².

4.2.2 Previous mental illness; central nervous system disease; liver disease, abnormal kidney function; abnormal heart and lung function; diabetes; American Society of Anesthesiologists (ASA) classification III or above.

4.2.3 Severe obstetric complications, such as preeclampsia and eclampsia, pregnancy-induced hypertension, placenta previa, or placental abruption.

4.2.4 Stillbirth, neonatal malformation.

4.2.5 Contraindications to epidural anesthesia, including abnormal coagulation function, taking anticoagulant therapy, severe hypovolemia, or hemodynamic instability.

4.3 Criteria of drop-out

4.3.1 Study intervention is not administered (change to general anesthesia) or protocol deviation occurs.

4.3.2 Intervention interrupted by the investigators/anesthesiologists (due to adverse events).

4.3.3 Use of a prohibited drugs.

Details of the above situations should be recorded and corrected when possible. The cases will be followed up according to the study protocol and included in the intention-to-treat analysis.

4.3.4 Withdraw consent after intervention started.

The situation should be recorded. The primary therapeutic effects recorded in the last time will be regarded as the final assessment results. The cases will be included in the intention-to-treat analysis.

4.4 Criteria of elimination

Enrolled patients will be excluded if they meet any of the following criteria:

151 4.4.1 Withdraw consents before intervention.

152 4.4.2 Surgery cancelled.

153 4.4.3 No assessment result of the primary outcomes.

154 The causes of elimination should be explained. The case will be excluded from the intention-
155 to-treat analysis. The case report forms will be preserved for reference.

156 ***4.5 Criteria of study interruption***

157 Study will be interrupted in the following situations:

158 4.5.1 Severe safety problem occurred during the study.

159 4.5.2 Serious mistake found in the protocol.

160 4.5.3 Fund or management problem of the investigators.

161 4.5.4 Study cancelled by the administrative authority.

162 Study interruption may be transient or permanent. All recorded case report forms will be
163 preserved for reference in case of study interruption.

165 **5. Randomization and Masking**

166 ***5.1 Randomization***

167 5.1.1 Random allocations will be generated using the SPSS 22.0 (IBM Corp, Armonk, NY,
168 USA) in a 1:1 ratio. Assignments will be concealed in sequentially numbered opaque envelopes.

169 5.1.2 For each participating center, a study coordinator will be designated to distribute the
170 randomization result to the anesthesiologists according to the sequence of recruited patients,
171 and to coordinate between investigators.

172 5.1.3 For each recruited patient, an anesthesiologist will be designated for anesthesia and data
173 collection according to the result of randomization.

174 5.1.4 Study intervention (use esketamine or normal saline) will be provided according to the
175 randomization results by anesthesiologists who do not participate in the outcome assessments.

176 ***5.2 Masking***

177 5.2.1 All patients, anesthesiologists, other health care team members, and investigators who
178 are responsible for data collection and follow-ups will be blinded to group allocation.

179 5.2.2 Investigators who are responsible for postoperative follow-up and outcome assessments
180 are not involved in anesthesia and perioperative management and have no knowledge of study
181 group assignment.

182 5.2.3 Statistical analysis will be performed by an independent statistician.

6. Intervention protocol

6.1 Anesthesia management

6.1.1 No premedication will be administered.

6.1.2 Intraoperative monitoring includes electrocardiogram (ECG) and oxygen saturation (SpO₂) and non-invasive blood pressure.

6.1.3 All patients will undergo caesarean section under epidural anesthesia and epidural puncture will be performed in the left lateral decubitus position. A 16-G Tuohy needle will be used for puncture in the L2-3 lumbar intervertebral space. After confirming the epidural space by the loss-of-resistance-to-air method, a catheter will be inserted 3-4 cm in a cephalad direction.

6.1.4 A test dose of 1.5% lidocaine 5 ml will be given. After a 5-minute observation, 0.75% ropivacaine 10 ml will be given, followed by an infusion of 0.75% ropivacaine at 3-5 ml/h.

6.1.5 Oxygen will be provided at 5 L·min⁻¹ via a face mask.

6.1.6 Study drugs will be administered when the targeted upper sensory block level was achieved (about 2 minutes before incision). Specifically, esketamine 0.25 mg·kg⁻¹ will be administered intravenously over 1 min for patients in the esketamine group; the same volume of normal saline will be injected for patients in the control group.

6.1.7 At the end of surgery, a patient-controlled epidural analgesia will be attached for postoperative analgesia, which is established with 0.2% ropivacaine and programmed to deliver a continuous infusion at 2 ml/h.

6.2 Remedial measures

6.2.1 If the epidural anesthesia is failed, spinal anesthesia or general anesthesia will be performed.

6.2.2 If the anesthesia level of the block is not sufficient, we will continue to add epidural drugs in fractions until the level of the block reaches at least T6.

6.2.3 The above measures will be recorded.

7. Data collection

7.1 Baseline data

7.1.1 Demographic parameters, pregestational comorbidity, number of gravidity and parturitions, duration of gestation, and ASA classification.

7.1.2 Pain intensity at rest will be assessed with the numeric rating scale (NRS; an 11-point scale where 0=no pain and 10=the worst pain) before anesthesia.

7.1.3 Maternal sedation level will be assessed with the Ramsay Sedation Scale (1=restlessness; 2=completely awake, quiet and cooperative; 3= drowsiness but responding to verbal commands;

4=light asleep but responding to touch or pain; 5= asleep but slowly responding to touch or pain; 6=deeply asleep and does not respond) before anesthesia.

7.2 Intraoperative data

7.2.1 Durations of anesthesia and surgery, fluid balance (including fluid infusion, estimated blood loss, and urine output), use of vasopressors.

7.2.2 Time intervals from study drug administration to neonatal delivery and from uterine incision to neonatal delivery.

7.2.3 Pain intensity at rest will be assessed with the NRS at the following timepoints: after anesthesia (before study drug administration), surgical incision (2 min after study drug administration), 5 min after study drug administration, immediately after fetal delivery (about 10 min after esketamine), and end of surgery.

7.2.4 Maternal sedation level will be assessed with the Ramsay Sedation Scale at the following timepoints: after anesthesia (before study drug administration), surgical incision (2 min after study drug administration), 5 min after study drug administration, immediately after fetal delivery (about 10 min after esketamine), and end of surgery.

7.2.5 Neonatal data including sex, birth weight, and Apgar scores at 1 and 5 minutes after delivery.

7.2.6 Immediately after childbirth, blood samples will be collected from artery of mothers and from umbilical artery and vein of fetuses in the esketamine group (in selected participants) and stored in a -20°C refrigerator. Esketamine concentrations were measured with the reverse phase high-performance liquid chromatography (RP-HPLC).

7.3 Postoperative data

7.3.1 Routine anesthetic follow-up after caesarean section.

7.3.2 Maternal psychiatric symptoms that may be caused by experimental drugs.

7.3.3 Pain intensity at rest will be assessed with the NRS at the following timepoints: 6 h after surgery, and 12 h after the surgery.

7.3.4 Maternal sedation level will be assessed with the Ramsay Sedation Scale at the following timepoints: 6 h after surgery, and 12 h after the surgery.

7.3.5 Admission to neonatal ward or neonatal intensive care unit.

7.3.6 Length of hospital stay after surgery.

8. Outcomes

8.1 Primary outcome

Our co-primary outcomes include maternal pain intensity and sedation score immediately after fetal delivery (about 10 min after esketamine). Pain intensity will be assessed with the NRS.

Maternal sedation level will be assessed with the Ramsay Sedation Scale.

8.2 Secondary outcomes

8.2.1 Maternal pain intensity as assessed with the NRS at the following timepoints: surgical incision (2 min after study drug administration), 5 min after study drug administration, end of surgery, 6 h after surgery, and 12 h after the surgery.

8.2.2 Maternal sedation level as assessed with the Ramsay Sedation Scale at the following timepoints: surgical incision (2 min after study drug administration), 5 min after study drug administration, end of surgery, 6 h after surgery, and 12 h after the surgery.

8.2.3 Neonatal Apgar score assessed at 1 and 5 minutes after birth.

8.2.4 Postnatal umbilical vein blood gas pH value.

8.3 Other outcomes

8.3.1 Mean blood pressure (MBP), heart rate (HR), and pulse oxygen saturation (SpO₂) measured before anesthesia, immediately after anesthesia, surgical incision (2 min after study drug administration), 5 min after study drug administration, immediately after fetal delivery (10 min after study drug administration), end of surgery, and 1 h after surgery.

8.3.2 The requirement of neonatal ward admission and length of hospital stay.

8.3.3 Plasma concentrations of esketamine in maternal blood, neonatal umbilical venous blood and umbilical arterial blood after birth.

9. Adverse events

9.1 Definition

An adverse event indicates any unpredictable, unfavorable medical event that is associated with any medical intervention and occurs during the study period. It can be related to the study intervention or otherwise. It can manifest as any uncomfortable signs (including abnormal laboratory findings), symptoms, or transient morbidity.

9.2 Predicted adverse events in this study

9.2.1 Hypotension was defined as systolic pressure decrease <20% of baseline.

9.2.2 Hypertension was defined as systolic pressure increase >20% of baseline.

9.2.3 Bradycardia was defined as heart rate <60 bpm.

9.2.4 Tachycardia was defined as heart rate >100 bpm.

9.2.5 Desaturation was defined as oxygen saturation <90%.

9.2.6 Nausea and vomiting.

9.2.7 Neurological and mental signs and symptoms (such as lethargy, diplopia, dizziness, headache, nightmare, hallucination, anxiety, and irritability) and nystagmus.

288 9.2.8 Adverse events will be managed according to routine practice.

289

290 **10. Severe adverse events**

291 **10.1 Definition**

292 A severe adverse event indicates any unpredictable medical events that lead to death, threat of
293 life, prolonged length of hospital stays, persistent disability or dysfunction, or other severe
294 event.

295 **10.2 Management**

296 In case of any severe adverse events, the study intervention will be stopped and treatment will
297 be initiated immediately.

298 **10.3 Record and report**

299 10.3.1 In case of any severe adverse event, apart from active treatment and record as above,
300 the principal investigator and the Ethics Committee will be informed within 24 hours in written
301 report.

302 10.3.2 In case of study intervention related death, immediately stop the clinical trial, report the
303 event to the Ethics Committee as soon as possible, record in detail and carefully preserve the
304 related documents.

305 10.3.3 Any severe adverse event must be followed up until it is completely resolved or when
306 therapy is ended.

307

308 **11. The rule of unmasking**

309 11.1 After the follow-up of all cases have been completed, the data of case report forms have
310 been checked as correct, and the data entry have been finished, a database inspection report
311 will be written by the data manager.

312 11.2 After the database is locked, unmasking will be conducted. And the database will be sent
313 to the statisticians for statistical analysis.

314

315 **12. Data management**

316 12.1 The investigators should record data timely, completely and correctly according to the
317 original observations and assessments.

318 12.2 The completed case report forms, after signed by the supervisors, will be sent to a clinical
319 data custodian.

320 12.3 After the data in the case report forms have been input and checked, the case report forms
321 will be stored in sequence order.

12.4 Data management will be inspected by Clinical Research Institute of Women's Hospital School of Medicine Zhejiang University.

13. Statistical analysis

Statistical analyses will be performed using IBM SPSS for Windows version 22.0 (IBM Corp, Armonk, NY, USA) and GraphPad Prism version 5.0 (GraphPad Software Inc, San Diego, CA, USA).

13.1 General principles

13.1.1 The primary and secondary outcomes will be analyzed in an intention-to-treat population, i.e., all patients are analyzed in the group to which they are randomized. Also, we will do per-protocol analysis for the primary endpoints after excluding patients with major protocol deviation.

13.1.2 For continuous variables, the Kolmogorov-Smirnov test will be applied to evaluate the distribution. Variables with normal distribution were presented as mean $\pm$ standard deviation (SD). Variables with non-normal distribution will be presented as median and interquartile range (IQR).

13.1.3 Categorical variables will be presented as number of cases (%).

13.1.4 For the primary outcomes, $p < 0.025$ ($0.05/2$) will be considered statistically significant. For other outcomes, two-tailed tests will be used and $p < 0.05$ will be considered to be statistically significant unless otherwise indicated after Bonferroni correction.

13.2 Patient recruitment and drop-out status

The status of patient recruitment and drop-out will be summarized and listed. Comparison of the overall elimination/drop-out rate between the two groups will be performed with Chi-Square test.

13.3 Demographics and baseline characteristics

13.3.1 Demographics and baseline data will be presented.

13.3.2 For between-group differences, numeric variables will be analyzed using independent-samples t test or Mann-Whitney U test; categorical variables will be analyzed using the chi-square test, continuity correction chi-square test or Fisher exact test.

13.4 Intra- and postoperative variables

Numeric variables will be analyzed using the independent-samples t tests or Mann-Whitney U tests; categorical variables will be analyzed using the chi-square tests, continuity correction chi-square tests or Fisher exact tests. Missing data will not be replaced.

13.5 Efficacy analysis

13.5.1 Evaluation of primary endpoints

For the primary outcomes (maternal pain intensity and sedation score immediately after fetal delivery), the differences between group will be analyzed using Mann-Whitney U test. Median differences and 95% CIs will be calculated with the Hodges-Lehmann estimators.

13.5.2 Evaluation of secondary and other endpoints

13.5.2.1 Discrete variables (maternal pain intensity and sedation score during the perinatal period, neonatal Apgar scores, and umbilical vein pH) will be analyzed with the Mann-Whitney U tests. Median differences and 95% CIs will be calculated with the Hodges-Lehmann estimators. Missing data will not be replaced.

13.5.2.2 Categorical variables (neonatal ward admission proportion) will be analyzed using the chi-square tests, continuity correction chi-square tests or Fisher exact tests. The rate differences and 95% CIs will be provided. Missing data will not be replaced.

13.5.2.3 Time-to-event variables (length of hospital stay after surgery) will be analyzed with the Kaplan-Meier estimators with differences between groups assessed by the log-rank test; Cox proportional hazards models will be used to calculate HRs and 95% CIs.

13.6 *Safety analysis*

13.5.1 Describe the occurrence of adverse events in each group.

13.5.2 Describe the management of adverse events when appropriate.

13.5.3 Describe the occurrence of severe adverse events.

13.5.4 The rates of adverse events and/or managements between the two groups will be compared with Chi-Square test, continuity correction Chi-Square test or Fisher exact test.

13.5.5 Missing data will not be replaced.

14. Quality control and quality assurance

14.1 *Training for investigators*

14.1.1 An investigator training program will be designed by the principal investigator. A study coordinator will be designated to organize and implement the training program, and to record and preserve the related documents.

14.1.2 Investigator training will be performed during the month before starting the study.

14.1.3 The training program will be repeated 1-2 times a year throughout the study period, or will be performed whenever necessary.

14.2 *Monitoring of study conduct*

14.2.1 The study will be monitored by the Women's Hospital School of Medicine Zhejiang University Clinical Research Institute.

14.2.2 A project specialist will be designated by the Women's Hospital School of Medicine Zhejiang University Clinical Research Institute and will verify that the conduct of the study,

392 the record of data and the analysis are in accord with the study protocol and related regulations.
393 Investigators should cooperate with the project specialist.

394 14.2.3 Before and during the study period, the project specialist will go to the study centers for
395 initiation inspection, regular inspection, and end of study inspection.

396 14.2.4 The contents of inspection include the following:

397 14.2.4.1 To verify that investigators are designated and completed the training program.

398 14.2.4.2 To verify the authenticity of participants, and the process to obtain written informed
399 consents.

400 14.2.4.3 To verify the eligibility of participants.

401 14.2.4.4 To verify the correctness of the randomization procedure.

402 14.2.4.5 To verify that the follow-ups and assessments are performed according to the study
403 protocol.

404 14.2.4.6 Original data will be inspected in at least 5% of the recruited participants. Original
405 data of the primary outcome will be inspected in 100% of the recruited participants.

406 14.2.4.7 To verify that all severe adverse events are reported to the Ethics Committee according
407 to the study protocol. The original data of all severe adverse events will be inspected.

408 14.2.4.8 To verify the transport, dissemination and retrieve of study drugs, and the records of
409 storage and return of study drugs.

410 14.2.4.9 To verify that the blood samples are collected and stored according to the study
411 protocol and the standard operating procedures.

412 14.2.4.10 To verify that the revised study protocol, participant-related documents, report of
413 severe adverse events, and annual summary report are submitted to the Ethics Committee
414 timely by the investigators for approval or record.

415 14.2.4.11 To verify the preservation of study-related documents and original data.

416 14.2.4.12 To verify the trial management in the study centers, the progress of participant
417 recruitment and the study conduct, the accomplishment of recruited cases, and the situation of
418 case drop-out.

419 14.2.5 A written report will be provided after each inspection.

420 **14.3 *Inspection of data quality***

421 14.3.1 The project specialist will check and verify the completeness and correctness of the data
422 recorded in the case report forms.

423 14.3.2 All data queries must be solved before the database can be locked for statistical analysis.

424 **15. Ethics requirements**

425 **15.1 *Ethics Committee***

The study protocol must be approved by the Women's Hospital School of Medicine Zhejiang University Institutional Review Board before the study can be started. The investigators must strictly follow the Helsinki Declaration and China's relevant clinical trial management regulations. The principal investigator is responsible to report the status and the progress of the study to the Institutional Review Board.

15.2 Written informed consent

For each potential participant, investigators are responsible to fully explain the purpose, procedures and possible risks of this study in a written form manner. The investigators must let every potential participant know that he/she has the right to withdraw consent from the study at any time. Every potential participant must be given a written informed consent. Every participant or the authorized surrogate of the participant must sign the consent before he/she can be enrolled in the study. The written informed consents will be kept as a part of the clinical trial documents.

15.3 Privacy and confidentiality

15.3.1 During the study period, the collected data from participants are labelled with special recruitment numbers and acronyms of names.

15.3.2 All personal information of the participants will be kept confidential.

15.3.3 Results of the study will be published as scientific articles. But all personal data (including name and age, etc.) are strictly confidential.

16. Study termination

16.1 In case that severe adverse events or serious quality problem occur during the study period, the study will be stopped. A report will be sent to the Ethics Committee. Restart of the study will need an approval from the Ethics Committee.

16.2 The study will be terminated after accomplishment of required patient recruitment and data collection. Decision will be made by the principal investigator.

17. Preservation of documents

Investigators will carefully preserve all documents and data of the clinical trial according to the requirements of Good Clinic Practice for a period of 5 years.

18. Declaration of interests

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Women's Hospital, School of Medicine, Zhejiang University (ZDFY2022-4XA102).

19. References

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Statistical analysis plan

This is a copy from the study protocol.

1. Study design and objectives

This is a multi-center, randomized, double blind trial with two parallel arms. The purpose of the study is to investigate the sedative and analgesic effects of intravenous esketamine before childbirth during cesarean delivery under epidural anesthesia.

2. Sample size estimation

Based on a pilot study, we suppose that the mean difference in NRS pain score immediately after fetal delivery (10 min after administering esketamine) would be 0.3 with standard deviations (SD) of 0.85 and 1.23, respectively, between the two groups. With alpha set at 5% and power at 90%, 263 patients will be needed in each group. Considering a drop-out rate of about 10%, we plan to enroll 300 patients in each group. The sample size was estimated with the PASS 15.0 software.

3. Efficacy outcomes

3.1 Primary outcome

Our co-primary outcomes include maternal pain intensity and sedation score immediately after fetal delivery (about 10 min after esketamine). Pain intensity will be assessed with the numeric rating scale (NRS; an 11-point scale where 0=no pain and 10=the worst pain). Maternal sedation level will be assessed with the Ramsay Sedation Scale (1=restlessness; 2=completely awake, quiet and cooperative; 3= drowsiness but responding to verbal commands; 4=light asleep but responding to touch or pain; 5= asleep but slowly responding to touch or pain; 6=deeply asleep and does not respond).

3.2 Secondary outcomes

3.2.1 Maternal pain intensity as assessed with the NRS at the following timepoints: surgical incision (2 min after study drug administration), 5 min after study drug administration, end of surgery, 6 h after surgery, and 12 h after the surgery.

3.2.2 Maternal sedation level as assessed with the Ramsay Sedation Scale at the following timepoints: surgical incision (2 min after study drug administration), 5 min after study drug administration, end of surgery, 6 h after surgery, and 12 h after the surgery.

3.2.3 Neonatal Apgar score assessed at 1 and 5 minutes after birth.

3.2.4 Postnatal umbilical vein blood gas pH value.

3.3 Other outcomes

3.3.1 Mean blood pressure (MBP), heart rate (HR), and pulse oxygen saturation (SpO₂) measured before anesthesia, immediately after anesthesia, surgical incision (2 min after study drug administration), 5 min after study drug administration, immediately after fetal delivery (10 min after study drug administration), end of surgery, and 1 h after surgery.

3.3.2 The requirement of neonatal ward admission and length of hospital stay.

3.3.3 Plasma concentrations of eskatamine in maternal blood, neonatal umbilical venous blood and umbilical arterial blood after birth.

4. Statistical analysis

4.1 General principles

4.1.1 The primary and secondary outcomes will be analyzed in an intention-to-treat population, i.e., all patients are analyzed in the group to which they are randomized. Also, we will do per-protocol analysis for the primary endpoints after patients with major protocol deviation.

4.1.2 For continuous variables, the Kolmogorov-Smirnov test will be applied to evaluate the distribution. Variables with normal distribution were presented as mean $\pm$ standard deviation (SD). Variables with non-normal distribution will be presented as median and interquartile range (IQR).

4.1.3 Categorical variables will be presented as number of cases (%).

4.1.4 For the primary outcomes, $p < 0.025$ ($0.05/2$) will be considered statistically significant. For other outcomes, two-tailed tests will be used and $p < 0.05$ will be considered to be statistically significant unless otherwise indicated after Bonferroni correction.

4.2 Patient recruitment and drop-out status

The status of patient recruitment and drop-out will be summarized and listed. Comparison of the overall elimination/drop-out rate between the two groups will be performed with Chi-Square test.

4.3 Demographics and baseline characteristics

4.3.1 Demographics and baseline data will be presented.

4.3.2 For between-group differences, numeric variables will be analyzed using independent-samples t test or Mann-Whitney U test; categorical variables will be analyzed using the chi-square test, continuity correction chi-square test or Fisher exact test.

4.4 Intra- and postoperative variables

Numeric variables will be analyzed using the independent-samples t tests or Mann-Whitney U tests; categorical variables will be analyzed using the chi-square tests, continuity correction

chi-square tests or Fisher exact tests. Missing data will not be replaced.

4.5 Efficacy analysis

4.5.1 Evaluation of primary endpoints

For the primary outcomes (maternal pain intensity and sedation score immediately after fetal delivery), the differences between group will be analyzed using Mann-Whitney U test. Median differences and 95% CIs will be calculated with the Hodges-Lehmann estimators.

4.5.2 Evaluation of secondary and other endpoints

4.5.1 Discrete variables (maternal pain intensity and sedation score during the perinatal period, neonatal Apgar scores, and umbilical vein pH) will be analyzed with the Mann-Whitney U tests. Median differences and 95% CIs will be calculated with the Hodges-Lehmann estimators. Missing data will not be replaced.

4.5.2 Categorical variables (neonatal ward admission proportion) will be analyzed using the chi-square tests, continuity correction chi-square tests or Fisher exact tests. The rate differences and 95% CIs will be provided. Missing data will not be replaced.

4.5.3 Time-to-event variables (length of hospital stay after surgery) will be analyzed with the Kaplan-Meier estimators with differences between groups assessed by the log-rank test; Cox proportional hazards models will be used to calculate HRs and 95% CIs.

4.6 Safety analysis

4.6.1 Describe the occurrence of adverse events in each group.

4.6.2 Describe the management of adverse events when appropriate.

4.6.3 Describe the occurrence of severe adverse events.

4.6.4 The rates of adverse events and/or managements between the two groups will be compared with Chi-Square test, continuity correction Chi-Square test or Fisher exact test.

4.6.5 Missing data will not be replaced.

Supplementary Online Content

Xu LL, Wang C, Deng C-M, et al. Efficacy and safety of esketamine for supplemental analgesia during elective cesarean delivery: a randomized clinical trial. *JAMA Netw Open*. 2023;6(4):e239321. doi:10.1001/jamanetworkopen.2023.9321

eTable 1. Enrollment by Study Site

eTable 2. Numeric Rating Scale of Pain and Ramsay Sedation Scale During the Perinatal Period

eTable 3. Blood Pressure, Heart Rate, and Pulse Oxygen Saturation During the Perinatal Period

eTable 4. Maternal and Fetal Esketamine Concentrations

eFigure. Comparison of the Mean Arterial Pressure (A), Heart Rate (B), and Pulse Oxygen Saturation (C) Between the Two Groups

This supplementary material has been provided by the authors to give readers additional information about their work.

eTable 1. Enrollment by Study Site

Study centers	City	Province	Number (%)
Women’s Hospital, Zhejiang University School of Medicine	Hangzhou	Zhejiang	180 (30.0%)
Jiangxi Maternal and Child Health Hospital	Nanchang	Jiangxi	160 (26.7%)
Hangzhou Women's Hospital	Hangzhou	Zhejiang	80 (13.3%)
Affiliated Xiaoshan Hospital, Hangzhou Normal University	Hangzhou	Zhejiang	120 (20.0%)
Jiaxing Maternity and Child Health Care Hospital	Jiaxing	Zhejiang	60 (10.0%)
Total			600 (100%)

eTable 2. Numeric Rating Scale of Pain and Ramsay Sedation Scale During the Perinatal Period

	Control group (n=300)	Esketamine group (n=300)	Median difference (95% CI) ^a	<i>p</i> value
Numeric rating scale of pain at rest, median (IQR), point ^b				
Before anesthesia	0 (0, 0)	0 (0, 0)	0 (0, 0)	0.430
After anesthesia (before study drug)	0 (0, 0)	0 (0, 0)	0 (0, 0)	0.672
Incision (2 min after study drug)	0 (0, 1)	0 (0, 1)	0 (0, 0)	0.258
5 min after study drug	0 (0, 1)	0 (0, 0)	0 (0, 0)	0.010
End of surgery	0 (0, 0)	0 (0, 0)	0 (0, 0)	0.296
6 h after surgery	1 (1, 2)	1 (0, 2)	0 (0, 0)	0.125
12 h after surgery	1 (1, 2) [2]	1 (1, 2) [2]	0 (0, 0)	0.472
Ramsay sedation scale, median (IQR), point ^c				
Before anesthesia	2 (2, 2)	2 (2, 2)	0 (0, 0)	0.254
After anesthesia (before study drug)	2 (2, 2)	2 (2, 2)	0 (0, 0)	0.103
Incision (2 min after study drug)	2 (2, 2)	3 (2, 4)	1 (1, 1)	<0.001
5 min after study drug	2 (2, 2)	4 (3, 5)	2 (2, 2)	<0.001
End of surgery	2 (2, 2)	2 (2, 2)	0 (0, 0)	0.477
6 h after surgery	2 (2, 2)	2 (2, 2)	0 (0, 0)	0.157
12 h after surgery	2 (2, 2) [2]	2 (2, 2) [2]	0 (0, 0)	0.083

p values in bold indicate <0.006 (0.05/8) and are considered statistically significant after Bonferroni correction. Numbers in square brackets indicate patients with missing data.

^a Calculated as the esketamine group minus the control group.

^b Score ranges from 0 to 10, where 0=no pain and 10=the worst pain.

^c Ramsay sedation score ranges from 1 (restlessness) to 6 (deeply asleep and does not respond) and 2 indicates completely awake, quiet and cooperative.

eTable 3. Blood Pressure, Heart Rate, and Pulse Oxygen Saturation During the Perinatal Period

	Control group (n=300)	Esketamine group (n=300)	Mean difference (95% CI) ^a	<i>p</i> value
Mean blood pressure, mean (SD), mmHg				
Before anesthesia	93 (11)	92 (11)	-0.9 (-2.6, 0.8)	0.311
After anesthesia (before study drug)	84 (12)	84 (13)	0.1 (-1.9, 2.1)	0.918
Incision (2 min after study drug)	82 (13)	85 (15)	3.2 (1.0, 5.5)	0.005
5 min after study drug	81 (12)	89 (15)	7.1 (5.0, 9.2)	<0.001
Immediately after fetal delivery	83 (12)	89 (12)	6.1 (4.2, 8.0)	<0.001
End of surgery	82 (11)	82 (11)	0.1 (-1.6, 2.0)	0.880
1 h after surgery	86 (10)	86 (11)	-0.4 (-2.1, 1.3)	0.644
Heart rate, mean (SD), bpm				
Before anesthesia	87 (14)	88 (13)	0.5 (-1.7, 2.7)	0.674
After anesthesia (before study drug)	90 (16)	91 (17)	0.8 (-1.8, 3.4)	0.545
Incision (2 min after study drug)	87 (17)	88 (19)	0.7 (-2.2, 3.5)	0.655
5 min after study drug	84 (16)	90 (17)	5.6 (3.0, 8.2)	<0.001
Immediately after fetal delivery	86 (16)	92 (15)	5.2 (2.8, 7.7)	<0.001
End of surgery	82 (13)	83 (12)	1.4 (-0.6, 3.4)	0.161
1 h after surgery	81 (12)	80 (11)	-0.4 (-2.3, 1.4)	0.662
Pulse oxygen saturation, mean (SD), %				
Before anesthesia	98.8 (1.1)	98.8 (1.2)	0.0 (-0.2, 0.2)	0.724
After anesthesia (before study drug)	99.2 (1.1)	99.2 (1.4)	0.0 (-0.2, 0.2)	0.869
Incision (2 min after study drug)	99.4 (1.0)	99.4 (1.1)	0.0 (-0.1, 0.2)	0.668
5 min after study drug	99.3 (0.9)	99.3 (1.4)	0.0 (-0.2, 0.2)	0.892
Immediately after fetal delivery	99.3 (0.9)	99.3 (1.3)	0.0 (-0.2, 0.1)	0.689
End of surgery	99.3 (0.9)	99.2 (1.0)	-0.1 (-0.2, 0.1)	0.476
1 h after surgery	98.7 (0.9)	98.7 (1.0)	0.0 (-0.2, 0.1)	0.864

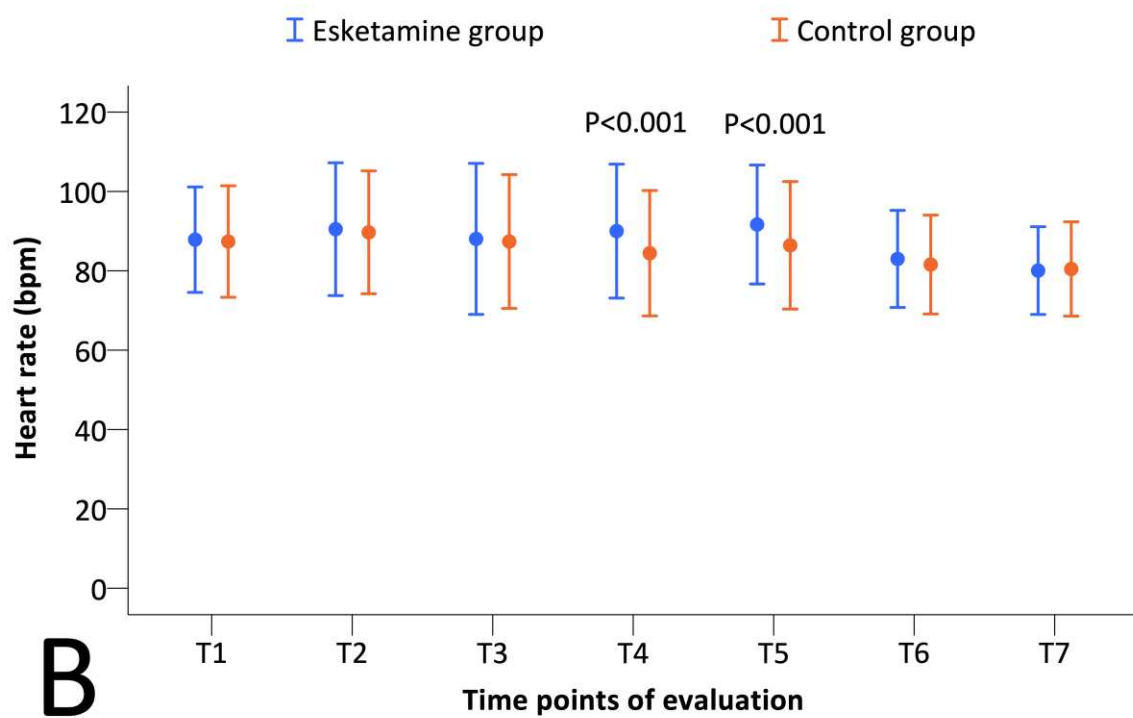
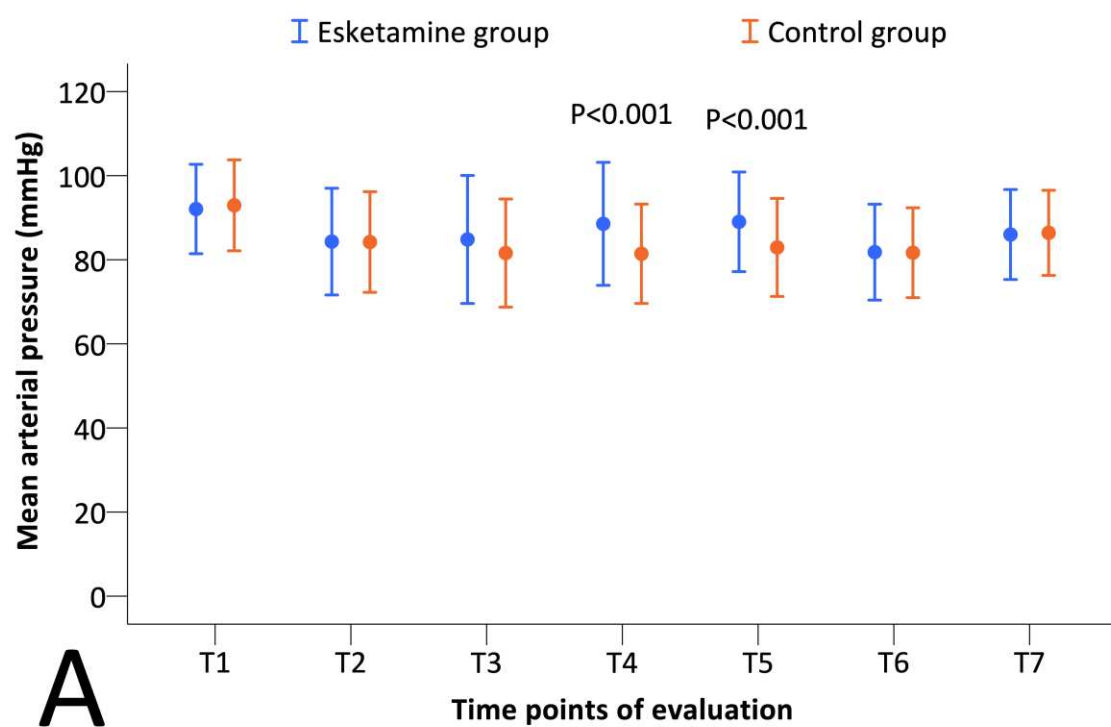
p values in bold indicate <0.007 (0.05/7) and are considered statistically significant after Bonferroni correction. Numbers in square brackets indicate patients with missing data.

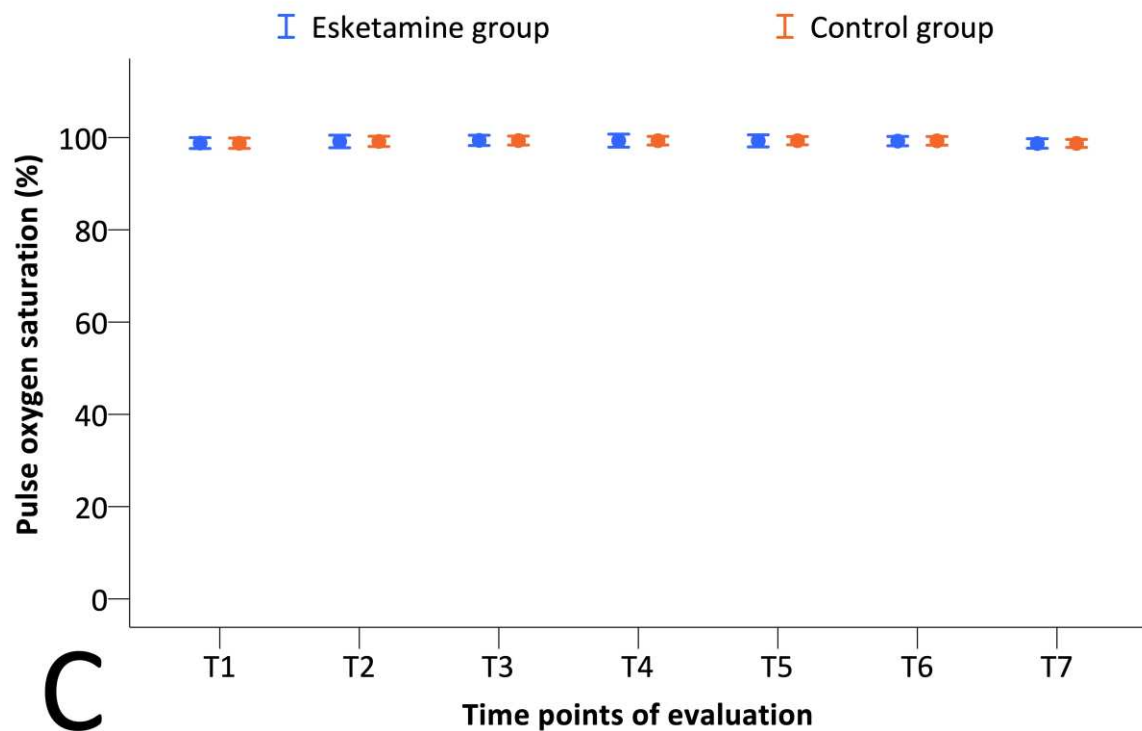
^a Calculated as the esketamine group minus the control group.

eTable 4. Maternal and Fetal Esketamine Concentrations

	Esketamine group (n=13)
Maternal artery, mean (SD), ng/ml	172.4 (79.8)
Fetal umbilical artery, mean (SD), ng/ml	62.0 (23.6)
Fetal umbilical vein, mean (SD), ng/ml	142.9 (66.7) ^a
Fetal umbilical vein/maternal artery, mean (SD)	0.83 (0.09)
Fetal umbilical artery/fetal umbilical vein, mean (SD)	0.48 (0.20)

^a $p=0.001$ when compared with the fetal umbilical artery concentration.





eFigure. Comparison of the Mean Arterial Pressure (A), Heart Rate (B), and Pulse Oxygen Saturation (C) Between the Two Groups

Data are mean and standard error. T1, before anesthesia; T2, immediately after anesthesia; T3, incision (2 min after study drug administration); T4, 5 min after study drug administration; T5, immediately after fetal delivery (10 min after study drug administration); T6, end of surgery; and T7, 1 h after surgery.

Data Sharing Statement

Xu. Efficacy and Safety of Esketamine for Supplemental Analgesia During Elective Cesarean Delivery. *JAMA Netw Open*. Published April 21, 2023.

doi:10.1001/jamanetworkopen.2023.9321

Data

Data available: Yes

Data types: Participant data with identifiers

How to access data: chenxinz@zju.edu.cn

When available: With publication

Supporting Documents

Document types: None

Additional Information

Who can access the data: researchers whose proposed use of the data has been approved

Types of analyses: for any purpose

Mechanisms of data availability: with investigator support

Any additional restrictions: no